

# Advanced Data Management

# Ontology-based Data Management

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# Outline of the course

1. Introduction to propositional logic
  2. Introduction to First-order Logic
  3. Relational Calculus
  4. Information Integration Systems (IIS)
  5. Logical formalization of IISs
  6. Mapping between **Global Schema** e **Data Sources**
  7. Incomplete information databases
  8. Query answering over IISs
  9. **Ontology-Based Data Management (OBDM)**
  10. Description Logic Ontologies
  11. Query answering in ontologies
  12. Query answering in OBDM
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# The Problem

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# Data management: a complex task

Extracting information and obtaining insights from data, as well as data governance are both challenging but necessary tasks

- the amount of data stored in current information systems continues to grow.
- proliferation of **heterogenous** and independent data sources both within a single organization and in cooperating environments.

We can easily understand why effectively accessing, **integrating**, and managing data in complex organizations is still **one of the main issues faced by the information technology (IT) today**

# Data Integration

Databases are great!

They let us manage efficiently huge amounts of data . . .

. . . assuming you have put all data into your schema.

Unfortunately, the reality is much more complicated and **heterogeneous**:

- Data sets were created independently.
- Data are often stored across different sources.
- Data sources are controlled by different people/organizations.

**Goal of data integration:** put together **different data sources**,

- created for **different purposes**,
  - and controlled by **different people**,
  - making them **accessible in a uniform way**.
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# Physical vs Conceptual Heterogeneity

- **Physical Heterogeneity:** Information is represented using different formats. Datatypes and data models of the information sources may differ.
  - **Conceptual Heterogeneity:** The information sources describe the world from different perspectives.
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# Is Heterogeneity the only problem?

Gathering information can be **difficult** even for **very simple information systems**

**EX:** This table contains information about the customers of a (real) organization, but...

| CUC     | TS_START    | TS_END      | ID_GRUP | FLAG_CP | FLAG_CF | FATTURATO | FLAG_FATT |
|---------|-------------|-------------|---------|---------|---------|-----------|-----------|
| 124589  | 30-lug-2004 | 1-gen-9999  | 92736   | S       | N       | 195000,00 | N         |
| 140904  | 15-mag-2001 | 15-giu-2005 | 35060   | N       | N       | 230600,00 | N         |
| 124589  | 5-mag-2001  | 30-lug-2004 | 92736   | N       | S       | 195000,00 | S         |
| -452901 | 13-mag-2001 | 27-lug-2004 | 92770   | S       | N       | 392000,00 | N         |
| 129008  | 10-mag-2001 | 1-gen-9999  | 62010   | N       | S       | 247000,00 | S         |
| -472900 | 10-mag-2001 | 1-gen-9999  | 62010   | S       | N       | 0 00      | N         |
| 130976  | 7-mag-2001  | 9-lug-2003  | 75680   |         |         |           |           |

# Is Heterogeneity the only problem?

...contains information about the customers of a (real) organization, but...

- **CUC** column contains the customer code: if it is positive, then it refers to an **ordinary** customer; if it is negative, to a **special** customer.
- **TS\_START** and **TS\_END** specify the time interval of validity for the record.
- **ID\_GROUP** indicates the group the customer belongs to (if the value of **FLAG\_CP** is “S,” then the customer is the **leader** of the **group**; if **FLAG\_CF** is “S,” then the customer is the **controller** of the group).
- **FATTURATO** is the annual turnover (it is valid only if **FLAG\_FATT** is “S”).

Obviously, each term such as **special**, **ordinary**, **leader**, etc. has a specific meaning in the company, and understanding such meaning is crucial for extracting **useful information**



# Data sources evolve in a chaotic way

Although the initial design of data sources in an organization might be adequate, corrective maintenance actions tend to reshape these sources into a form that **diverges from the original structure**.

Data sources are often subject to **changes** to be adapted to new needs and guaranteeing their seamless usage within the organization is costly.

The result is that the data are stored in different sources and tend to be redundant, mutually inconsistent, and **obscure** for large classes of users.

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## A new paradigm

The preceding examples and observations show that it is not only important to provide a **unified access** to data, but that it is also critical to know how the data are **organized** and how to **interpret** them (their semantics).

A promising direction for addressing these challenges is provided by the **Ontology-Base Data Integration (OBDI)** paradigm.

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# Ontology-based Data Management

**Ontology-based Data Management (OBDM)** is a semantic paradigm, rooted on the idea of using suitable techniques from the area of **knowledge representation and reasoning** in AI for a new way to achieve data governance and integration, based on the principle of managing heterogeneous data through the lens of an **ontology**.

It is characterized by the following principles:

- Data may reside where they are (virtual approach)
- Build a conceptual specification of the domain of interest, in terms of knowledge structures (the ontology)
- Map such knowledge structures to concrete data sources
- Express the user queries over the knowledge structures
- Automatically translate the queries over the data sources

# What is an ontology?

We can define an **ontology** as an explicit specification of a conceptualization, i.e., an **abstract, simplified** view of the world that we wish to represent.

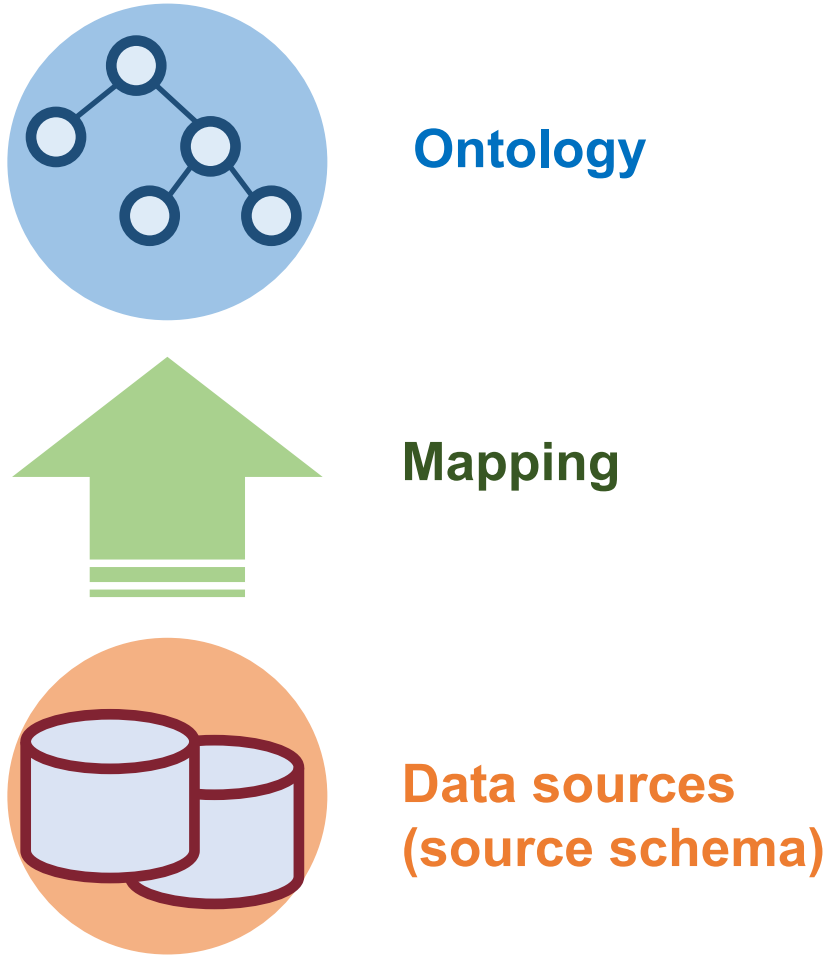
In other words, an ontology is a description of the **concepts** and **relationships** that can exist in a specific domain of interest for an agent or a community of agents.

In AI, an **ontology** provides a set of definitions that associate a name to the entities in a domain (e.g., classes, relations, individuals, etc.), and **formal axioms** that describe the relations between such entities.

Formally, an ontology is a **logical theory**.

The most common languages used for expressing ontologies are **Description Logics** and **OWL** (Web Ontology Language), the W3C standard language for ontologies.

# Ontology-based Data Management: architecture



Based on three main components:

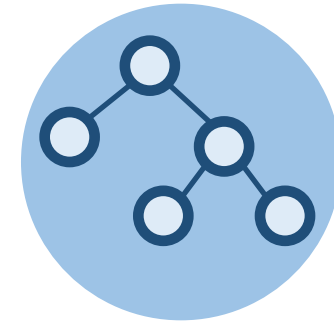
- The **Ontology** provides a formal conceptualization of a domain of interest
- The **Data sources**, representing external, independent, heterogeneous, data structures
- The **Mappings**, species the relationship between **ontology** elements and data at the **sources**

Users access only the conceptual layer, while the data layer, hidden to the clients, manages the data and remain autonomous

# OBDM and Information Integration

OBDM can be seen as a sophisticated form of **information integration** where the usual **global schema** is replaced by the conceptual model of the application domain, formulated as an **ontology**.

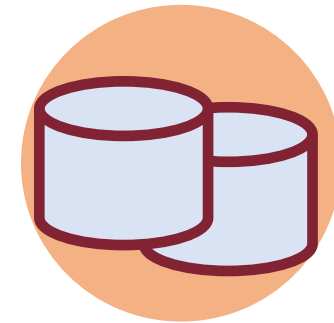
The integrated view is not anymore merely a data structure accommodating the various data at the **sources**, but a **semantically rich description** of the concepts of the domain and the relationships between them.



**Ontology**

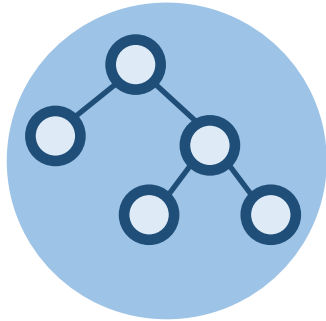


**Mapping**



**Data sources  
(source schema)**

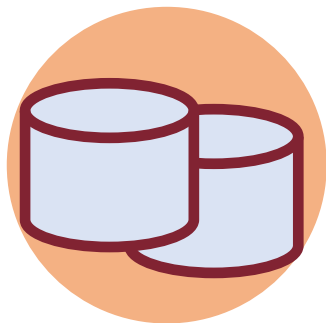
# OBDM vs Data Integration



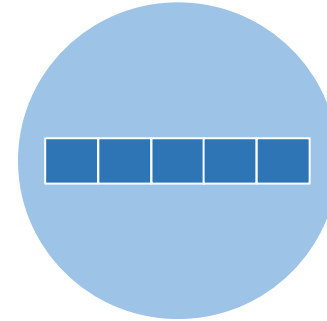
**Ontology**



**Mapping**



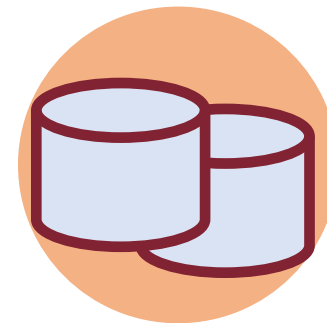
**Data sources  
(source schema)**



**Global Schema**



**Mapping**



**Data sources  
(source schema)**

## Credits

The presentations of this module of the course are based on the original slides of Prof. Giuseppe De Giacomo, Prof. Diego Calvanese, Prof. Marco Console, Prof. Maurizio Lenzerini, and Prof. Domenico Lembo.